START HERE – Data Analytics, Data Engineering & Data Science Course Package

This folder contains everything needed to build and record the 8-hour beginner course **"Data Analytics, Data Engineering & Data Science – Zero to First Project"** (11 modules, 55 videos, 480 minutes). It covers all three data roles at an introductory level: analytics (Modules 2–4, 6, 8), engineering (Module 4 databases and the Module 7 pipeline) and science (Modules 3 and 9). It contains:

- the project data
- every module's theory notes, word-for-word scripts and slide decks
- the code for the ShopKart project, built module by module
- practice tasks, quizzes, the final exam and the capstone
- the certificate

Read these first (in this order)

1. `01_Course_Guide/2_ShopKart_Project_Roadmap.docx`: how the single ShopKart project runs through all 11 modules. It covers what each module teaches, what we build, the goal, where coding starts, and which files pass from one module to the next.
2. `01_Course_Guide/1_Course_Overview.docx`: the course map, timings, format codes and certificate rules.
3. `01_Course_Guide/3_Tools_and_Google_Colab_Guide.docx`: which tool is used where, when Colab is used, and how to open and run each notebook.
4. `01_Course_Guide/4_Prep_and_Recording_Plan.docx`: the order to complete the project yourself before recording, then how to record.

Folder map

Folder	What's inside
`01_Course_Guide/`	The four guides above and the project flow diagram
`02_Datasets/`	ShopKart (course project) and FitGym (learner capstone) data, plus `DATA_DICTIONARY.docx`
`03_Modules/Module_01_…` to `Module_11_…`	One folder per module (see below)
`04_Assessments/`	Final exam pool (126 Q), exam rules, FitGym capstone brief, rubric, reference solution, certificate template and process
`05_Cheat_Sheets/`	Printable PDFs: Excel/Sheets, SQL, Python basics, pandas, data engineering, chart chooser, ML basics

Inside every module folder

File	What it's for
`1_Module_Guide.docx`	**For you.** The goal, where the module sits in the

File	What it's for
	project, input and output files, what we teach, what we build step by step, where the coding happens, a do-it-yourself checklist with the numbers you should get, and common learner mistakes
`2_Theory_Notes.docx`	The full explanation of every concept with ShopKart examples. Teach from it and hand it to learners as notes
`3_Video_Scripts.docx`	For every video: a timed shot plan and the word-for-word script. [Brackets] mark on-screen actions and slide numbers
`4_Slides_ModuleXX.pptx`	The PowerPoint deck. Sections follow the videos, and speaker notes give the cue for each slide. "Handwriting board" slides list what to draw
`5_Quiz.docx` / `5_Quiz.csv`	The module quiz with answers and explanations. The CSV is for uploading to your course platform
`project/`	The ShopKart project files for this module: Excel workbooks (2–3), `.sql` files (4, 10), Colab notebooks (5–10), and the pipeline script `shopkart_pipeline.py` (7). `*_follow_along` has blanks for recording; `*_solution` is complete and already run
`practice/`	Practice tasks for learners, with answers

Where the coding starts

- **Modules 1–3:** no code. Spreadsheet clicks and formulas only.
- **Module 4:** the first code (SQL in the browser).
- **Module 5:** Python starts in Google Colab.
- **Modules 6–10:** pandas, the data pipeline (Module 7), charts, machine learning and the capstone, all in Colab.

Checked before delivery

- Every Colab solution notebook runs top to bottom without errors.
- Every SQL query was run against the database, and its real result is written under it.
- Every number in the scripts and slides comes from the data.
- Modules 2 (Excel), 4 (SQL), 6 (pandas) and 7 (pipeline) give the same totals. For example, revenue is ₹1,47,04,649.70 in all four, and the Module 7 pipeline rebuilds `shopkart_clean.csv` identical to Module 6's.

Before you go live

- Add fresher salary ranges to the Module 1 job-titles slide only after checking current job portals.
- Do one dry run of the Power BI steps (`Module_08_Visualization/project/M08_8.4_PowerBI_Dashboard_Build_Guide.docx`). They have not been tested in the app.
- Fill in the certificate placeholders (verification domain, signature).

- Give learners the follow-along and practice-question files only. Keep the solutions and the FitGym reference solution for yourself and your graders.